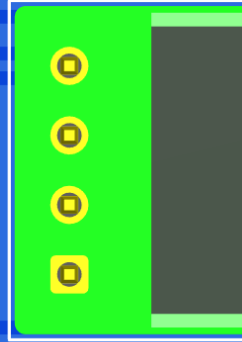
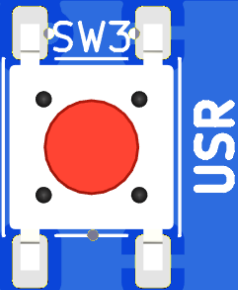
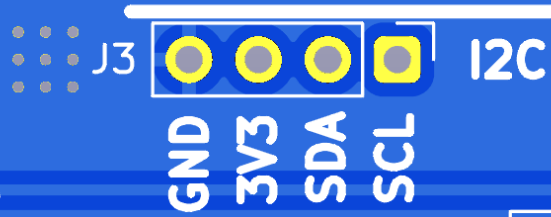
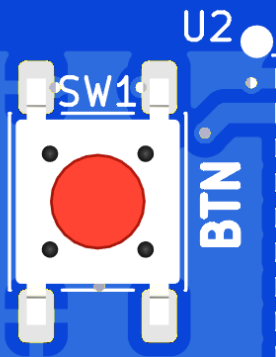


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Communication Hardware in Robotics

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Objectives

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Introduction to communication hardware components

3

Microcontrollers, microprocessors, and communication interfaces

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Microcontrollers, microprocessors, and communication interfaces

4

Integration of communication hardware in robotic systems

Introduction to communication hardware components

In robotics, communication hardware components play a crucial role in enabling various parts of a robotic system to exchange information.

Key communication hardware components commonly used in robotics:

- Microcontrollers and Processors
- Communication Modules
- Wireless Communication Modules
- Network Interfaces
- Sensors and Actuators Communication
- Human-Robot Interaction Interfaces
- Power Supplies and Battery Management Systems
- Communication Hubs and Gateways

Microcontrollers, microprocessors, and communication interfaces

- **Microcontrollers** (e.g., Arduino, PIC, AVR)
- and
- **Processors** (e.g., Raspberry Pi, Intel NUC) are central to communication.
- They handle data processing and control tasks, and often include built-in communication interfaces.

Actuator Control

Actuator controllers manage the operation of actuators, which are devices responsible for moving or controlling a mechanism or system in the robot.

Actuators include **motors, servos, hydraulic cylinders, and pneumatic actuators.**

- **Motor Controllers**
- **Servo Controllers**
- **Step/Dir Motor Controller**
- **Hydraulic Controllers**

Motor & Servo Controllers

- **PWM** (Pulse Width Modulation): Common method to vary motor speed.
- **Serial** (UART, SPI, I2C): Allows microcontrollers or computers to send commands and receive feedback.
- **CAN Bus** (Controller Area Network): Popular in robotics for real-time communication with multiple motor controllers.

Step/Dir Motor Controllers

- **Step/Dir Interface:** A pair of digital pins, where one controls direction and the other sends step pulses.
- **GPIO:** Often used with microcontrollers

Feedback Mechanisms

Feedback mechanisms provide information about the actual state of the actuators and the robot's performance. This feedback is essential for achieving accurate control and maintaining the desired behavior of the robot.

- Encoders
- Resolvers
- Potentiometers
- Pressure Sensors
- F/T sensors
- Accelerometers/Gyroscopes

Encoders

- **Purpose:**
 - Measure rotational position, velocity, or displacement of motors or joints.
- **Types:**
 - **Incremental:** Output pulses as the shaft rotates.
 - **Absolute:** Provide a unique code for each shaft position.
- **Communication:**
 - **Digital Output (A/B Channels):** Common in incremental encoders.
 - **SSI (Synchronous Serial Interface) or BiSS:** Used in high-res absolute encoders.
 - **I2C/SPI:** Found in modern compact encoders for embedded systems.

Resolvers

- **Purpose:**
 - Analog position sensors used in harsh environments (like industrial robotics).
- **Communication:**
 - Analog Signal Output: Sinusoidal signals represent angle.
 - Requires Resolver-to-Digital Converter (RDC) to digitize signals for controller use.
 - Often integrated via SPI or parallel bus post-conversion.

Potentiometers

- **Purpose:** Measure angular or linear displacement.
- **Communication:**
 - Analog Voltage Output: Proportional to position.
 - Read via ADC (Analog-to-Digital Converter) on a microcontroller or controller board.

Pressure Sensors

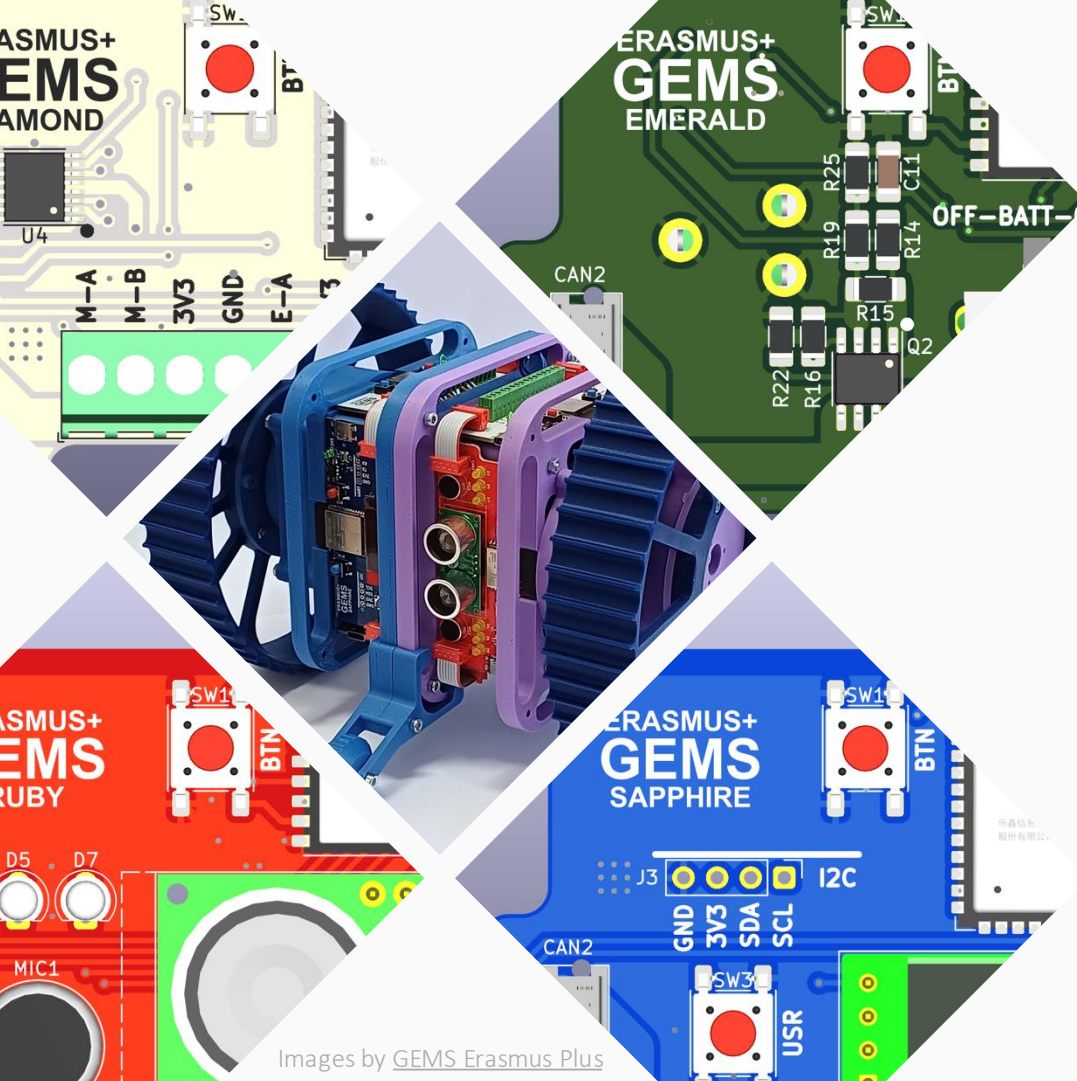
- Purpose: Monitor pressure in pneumatic or hydraulic actuators.
- Communication:
 - Analog Output (0–5V or 4–20mA): Traditional industrial standard.
 - I2C/SPI/UART: Available in digital smart sensors.
 - Often integrated with safety systems or force feedback loops.

F/T Sensors (Force/Torque Sensors)

- **Purpose:** Measure forces and torques acting on robotic joints or end-effectors.
- **Communication:**
 - EtherCAT / CAN / Ethernet: Real-time communication in high-performance robots.
 - USB or Serial: Used in research and desktop robots.
 - Commonly used in compliance control, haptics, and manipulation tasks.

Accelerometers / Gyroscopes (IMUs)

- **Purpose:** Provide orientation, acceleration, and angular velocity data.
- **Communication:**
 - **I2C/SPI/UART:** Standard in digital MEMS-based IMUs.
 - May output **quaternion**, **Euler angles**, or **raw sensor data**.
 - Often fused with magnetometers in a **9-DoF IMU** for accurate motion tracking.



Images by [GEMS Erasmus Plus](#)

Thank you for watching!

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